

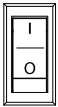
Troubleshooting guide

This guide provides the most common causes and solutions.

- Before troubleshooting, study the *Powermax30 AIR schematic* on page 230 and understand the *Theory of operation* on page 57.
- Before purchasing any major replacement component, verify the problem with Hypertherm Technical Service or the nearest Hypertherm repair facility listed in the front of this manual.
- See *System tests* on page 77 for detailed test procedures.



Power LED faults

  The ON/OFF power switch is set to ON (I), but the power ON LED does not illuminate.	
Possible causes	Possible solutions
❑ The system has no incoming voltage or an improper incoming voltage. ❑ The power cord, power switch, or power board is faulty.	❑ Make sure the electrical power is connected to an appropriately sized circuit. ❑ Make sure the system did not trip the circuit breaker. ❑ Perform <i>Test 1 – voltage input</i> on page 82.

 The power ON LED illuminates, but no fault LEDs illuminate, and no gas flows when you pull the torch trigger.	
Possible causes	Possible solutions
❑ The start signal is not reaching the control board due to: ● Damaged torch or torch lead ● Faulty power board OR ● Faulty control board	❑ Inspect the torch and torch lead, and replace if necessary. ❑ Make sure the control board start LED illuminates when you pull the torch trigger. If it does not, perform <i>Test 6 – plasma start</i> on page 92.



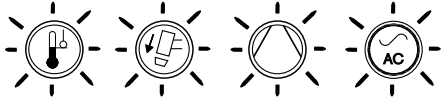
The power ON LED blinks.

Possible cause

- ❑ The incoming voltage is below 90 VAC or above 285 VAC.

Possible solution

- ❑ Perform *Test 1 – voltage input* on page 82.



All four LEDs blink when the system is powered ON.

Possible causes

- ❑ A major fault has occurred in the power supply. For example, the fan, solenoid valve, control board, or power board may be faulty.

Possible solutions

- ❑ The error LED on the control board should be blinking. The number of times it blinks between pauses indicates which components to test. See *Control board LEDs* on page 75.



Temperature LED faults

 The temperature LED illuminates while the machine is powered ON.	
Possible causes	Possible solutions
❑ The system overheated because the duty cycle was exceeded. ❑ The system is too cold to operate. The ambient temperature is too low.	❑ Let the system cool before using it. Leave the system on to allow the fan to cool the power supply. The fan runs for 7 minutes after postflow. See <i>Understand duty-cycle limitations</i> on page 50. ❑ Perform <i>Test 8 – fan</i> on page 95. ❑ Let the system warm up before using it. The operating temperature range for this system is -10°C (14°F) to 40°C (104°F).

 The temperature LED blinks while the machine is powered ON.	
Possible causes	Possible solutions
This fault occurs when the system continuously draws too much input current for too long. The fault protects the power switch and power cord from damage caused by overheating. This fault most often occurs when: ❑ You are operating the system on a 120 VAC input circuit and the cutting current is set above 20 A. ❑ You are operating the system on a 120 V / 15 A circuit. ❑ You are operating the system on a 120 VAC input circuit and are frequently stretching the arc while cutting. ❑ You are using an extension cord that is too long. The maximum length for an extension cord on a 120 VAC input circuit is 16 m (53 feet). The maximum length for an extension cord on a 240 VAC input circuit is 40.5 m (133 feet).	Let the system cool for 3 minutes before using it. Leave the system on to allow the fan to cool the power supply. To prevent the fault from occurring: ❑ Operate the system on a 240 VAC input circuit whenever possible. ❑ Turn down the cutting current. See <i>Step 3 – Adjust the output current</i> on page 47. ❑ If you do operate the system on a 120 VAC input circuit, use a 20 A circuit and set the cutting current below 20 A. ❑ Avoid stretching the arc. Drag the torch on the workpiece. See <i>Edge start on a workpiece</i> on page 54. ❑ Operate the system without using an extension cord. If you must use an extension cord, use a heavy conductor cord of the shortest possible length. See <i>Extension cord recommendations</i> on page 33.



Internal compressor LED faults



The internal compressor LED never illuminates by itself. It always illuminates with the temperature LED.



The internal compressor LED and the temperature LED illuminate while the machine is powered ON and the torch trigger is pulled.

Possible causes

- ❑ The internal compressor overheated and turned off. You may have exceeded the duty cycle for the system. See *Understand duty-cycle limitations* on page 50.

Possible solutions

- 1. Allow the power supply to cool for 4 minutes. Leave the system on to allow the fan to cool the power supply. Then try to produce an arc by pulling the torch trigger.**

When the internal compressor overheats, the compressor turns off, and the compressor and temperature LEDs remain illuminated for 4 minutes.

The system does not prevent you from firing the torch before the 4 minutes elapses. However, if you begin cutting before the compressor has time to fully cool down, the same fault is likely to occur again.

If the fault LEDs persist, continue with the following troubleshooting steps in the order listed.

- ❑ Environmental conditions or worn consumables are adversely affecting the power supply's performance.

- 2.** Make sure the louvers in the cover are not blocked. Blocking the louvers in the cover can prevent proper air circulation needed to cool internal components.
- 3.** Place the power supply right-side up on a level surface. Do NOT lay the power supply on its side. Laying the power supply on its side can cause the air filter to divert air away from the torch.
- 4.** Inspect and replace the consumables if they are worn or damaged. Always replace the nozzle and electrode together.
- 5.** Was the system stored in temperatures below freezing? If so, ice might have formed in the air-flow system inside the power supply. Move the power supply to a warmer location to allow the ice to melt.

  The internal compressor LED and the temperature LED illuminate while the machine is powered ON and the torch trigger is pulled.	
Possible causes	Possible solutions
☐ The air inlet filter on the internal compressor is completely clogged. For the compressor's air inlet filter to trigger this LED fault, the filter would have to be so blocked that no air can flow through it.	6. Replace the compressor's air inlet filter. See <i>Replace the air inlet filter in the internal compressor</i> on page 174.
☐ One of the following internal components is faulty and needs to be replaced: • Internal compressor • Compressor-driver board • Power board	7. Systematically check the internal compressor, the compressor-driver board, and the power board to determine which component needs to be replaced. Perform <i>Test 9 – internal compressor, compressor-driver board, and power board</i> on page 96.

    The internal compressor LED and the temperature LED blink alternately when the machine is powered ON.	
Possible causes	Possible solutions
☐ The system was powered on with the torch trigger being pulled. ☐ The start circuit is stuck closed.	☐ Release the torch trigger and reset the system by turning it OFF (O) and then turning it ON (I) again. The system automatically disables itself when the power supply is turned on while the torch trigger is pulled. ☐ Perform <i>Test 6 – plasma start</i> on page 92.



Torch cap LED faults



The torch cap LED illuminates while the machine is powered ON.

Possible causes

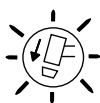
- ❑ The cap-sensing circuit is open due to:
 - The consumables are loose, incorrectly installed, or missing.

OR

 - The cap-sensor switch, torch lead, or torch body is faulty.

Possible solutions

- ❑ Make sure the consumables are installed correctly.
- ❑ Perform *Test 7 – torch cap-sensor* on page 94.



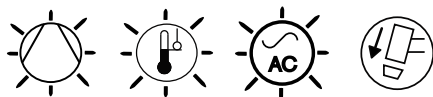
The torch cap LED blinks while the machine is powered ON.

Possible causes

- ❑ The consumables are stuck in an open or closed position because they are installed incorrectly, worn, or damaged.
- ❑ The nozzle and electrode are not touching when the torch trigger is pulled. The torch plunger is stuck.
- ❑ The torch or torch lead has a broken or short-circuited wire.

Possible solutions

- ❑ Inspect the torch lead. If it is twisted or kinked, straighten it out and try to fire the torch again.
- ❑ Make sure the consumables are installed correctly. See *Step 1 – Install the consumables* on page 45.
- ❑ Inspect the consumables for wear, and replace if necessary. Always replace the nozzle and electrode together.
- ❑ Perform *Test 5 – torch stuck open or torch stuck closed* on page 90.



The internal compressor, temperature, and power ON LEDs blink, and the torch cap LED illuminates.

Possible causes

- ❑ The torch was repeatedly fired with worn out consumables.
- ❑ The inverter is saturated (is in an over-current condition).

Possible solutions

- ❑ Install new consumables in the torch (they may be corroded or approaching end of life).
- ❑ If you continue to see this error, replace the power board. See *Replace the power board* on page 135.

Common cutting issues

Problem	Possible causes	Possible solutions
When you pull the torch trigger: - Gas flows from the torch, but the torch does not fire or fires only for a short period of time. OR - The pilot arc starts but then extinguishes before the normal 5-second timeout period.	- The consumables are worn or damaged. - The torch or torch lead is damaged. - The air filter element is dirty. - There is a voltage imbalance on the power board. The power board may be faulty.	- Inspect the consumables, torch, and torch lead, and replace if necessary. - Inspect the air filter element, and replace if necessary. See <i>Replace the air filter bowl and air filter element</i> on page 160. - Perform <i>Test 3 – VBUS and voltage balance on power board</i> on page 86.
The arc goes out during cutting or intermittently will not fire.	The work lead connection is poor. The work lead may be damaged or not properly connected to the workpiece.	- Inspect for loose connections at the ground clamp and at the power supply, and repair if necessary. - Reposition the work lead on the workpiece. - Clean the cutting surface to make a better connection with the work lead.
The pilot arc extinguishes when you move the plasma arc off the workpiece while still pulling the torch trigger.	- The consumables are worn or damaged. - The system is connected to a generator and is running on low input voltage (for example, less than 100 V). - The continuous pilot arc is not functioning correctly. The power board may be faulty.	- Inspect the consumables and replace if necessary. - If you use a generator, make sure it is properly rated for this system. See <i>Generator recommendations</i> on page 33. - Perform <i>Test 2 – power board voltage checks</i> on page 84 and <i>Test 3 – VBUS and voltage balance on power board</i> on page 86.

Problem	Possible causes	Possible solutions
Moisture exits the torch nozzle.	▪ A small amount of moisture at the torch nozzle is normal for this system, particularly in very humid environments. ▪ A larger amount of moisture at the torch nozzle may indicate that the air filter is not properly removing moisture from the system.	▪ A small amount of moisture should not prevent the torch from firing or degrade cut quality or performance. ▪ In case of excess moisture, examine the air filter inside the power supply. It is normal to see water in the filter bowl. The filter purges water when the water level in the bowl is high enough to engage the float mechanism. If the water level in the filter bowl is approximately half full (or more), the air filter may not be purging water properly. ▪ Make sure the drain hose connected to the bottom of the air filter bowl is not twisted or clogged. ▪ Water purged from the air filter exits through a hole in bottom of the of the power supply. Make sure this hole is not blocked. ▪ Replace the air filter bowl. See <i>Replace the air filter bowl and air filter element</i> on page 160. ▪ If the problem persists, replace the whole air filter assembly. See <i>Replace the air filter assembly</i> on page 157.

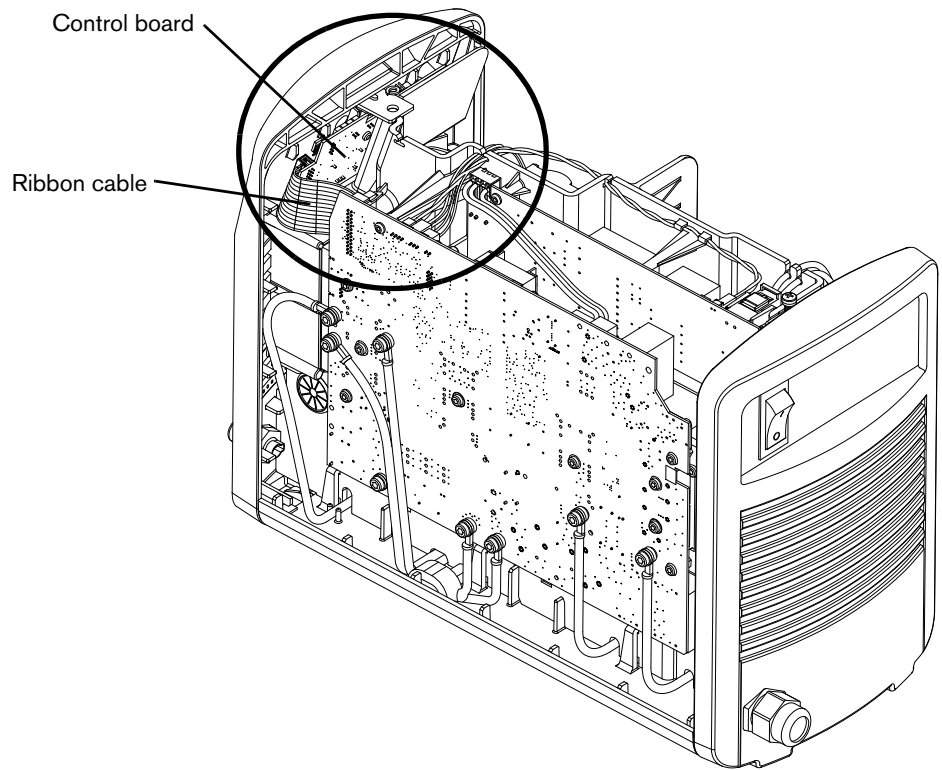
Problem	Possible causes	Possible solutions
The arc blows out but re-ignites when the torch trigger is pulled again, or the arc sputters and hisses.	- Too much moisture is present in the consumables. - The consumables are worn or damaged.	- Inspect the consumables for signs of excess moisture. A small amount of moisture at the torch nozzle is normal for this system and should not prevent the torch from firing. If excess moisture is present, fire the torch approximately 5 times in succession. Fire the torch in the air, not on the workpiece. - If the problem persists and excess moisture is not an issue, inspect and replace the consumable parts if they are worn or damaged. - If firing the torch does not clear out the excess moisture, examine the air filter inside the power supply. It is normal to see water in the filter bowl. The filter purges water when the water level in the bowl is high enough to engage the float mechanism. If the water level in the filter bowl is approximately half full (or more), the air filter may not be purging water properly. - Make sure the drain hose connected to the bottom of the air filter bowl is not twisted or clogged. - Water purged from the air filter exits through a hole in bottom of the of the power supply. Make sure this hole is not blocked. - Replace the air filter bowl. See <i>Replace the air filter bowl and air filter element</i> on page 160. - If the problem persists, replace the whole air filter assembly. See <i>Replace the air filter assembly</i> on page 157.

Problem	Possible causes	Possible solutions
The cut quality is poor, or the cut does not sever the metal.	▪ The consumables are worn or damaged. ▪ The work lead connection is poor. The work lead may be damaged or not properly connected to the workpiece. ▪ The amperage adjustment knob is set too low. ▪ The air pressure to the torch is too low. ▪ The power board is producing low current and may be faulty. ▪ The extension cord is too long, is damaged, or is not capable of delivering sufficient power to the power supply.	▪ Inspect the consumables and replace if necessary. ▪ Reposition the work lead on the workpiece. ▪ Clean the workpiece surface to make a better connection with the work lead. ▪ Inspect the work lead and replace if necessary. ▪ If your input circuit is capable, turn the amperage adjustment knob to a higher setting. ▪ Check the gas supply line for leaks or defective parts. Perform <i>Test 10 – check air pressure</i> on page 101. ▪ Perform <i>Test 2 – power board voltage checks</i> on page 84 and <i>Test 3 – VBUS and voltage balance on power board</i> on page 86. ▪ Operate the system without using an extension cord. If you must use an extension cord, use a heavy conductor cord of the shortest possible length. See <i>Extension cord recommendations</i> on page 33.
The circuit breaker trips frequently while you are cutting.	▪ The amperage adjustment knob is set too high for the input circuit. ▪ The wrong consumables are being used. ▪ The input circuit is not delivering sufficient power to the power supply. ▪ The operator is stretching the arc by holding the torch too far from the workpiece when cutting. ▪ The extension cord is too long, is damaged, or is not capable of delivering sufficient power to the power supply.	▪ Turn down the cutting current. See <i>Step 3 – Adjust the output current</i> on page 47. ▪ Make sure the correct consumables are installed for this power supply and torch. See <i>Consumable use</i> on page 37. ▪ Operate the system on a 240 VAC input circuit whenever possible. ▪ Avoid stretching the arc. Drag the torch on the workpiece. See <i>Edge start on a workpiece</i> on page 54. ▪ Operate the system without using an extension cord. If you must use an extension cord, use a heavy conductor cord of the shortest possible length. See <i>Extension cord recommendations</i> on page 33. ▪ Make sure nothing else is drawing power on the same circuit.

Control board LEDs

The control board (PCB1) is located inside of the front panel.

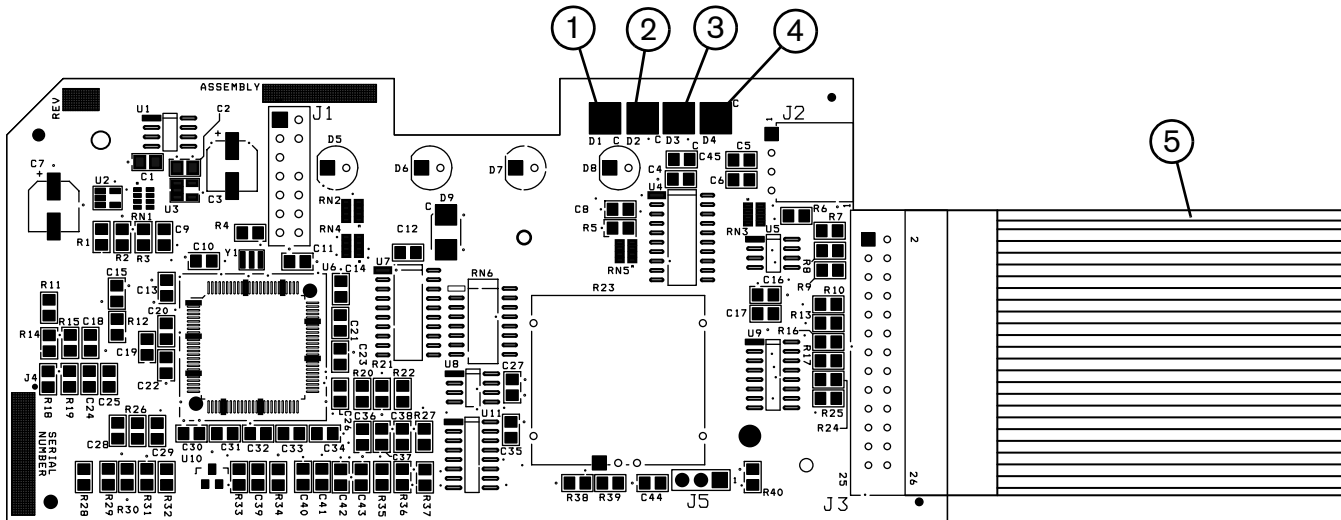
Figure 3



The control board has four diagnostic LEDs:

- **Reset** – This LED illuminates when certain voltage readings to the power board are out of range. See *Reset LED* on page 76.
- **Error** – This LED illuminates when the internal compressor, torch cap, or temperature LEDs on the front of the power supply illuminate. If all four LEDs on the front of the power supply are blinking, the Error LED also blinks. The number of blinks between pauses indicates which component may have failed.
- **Transfer** – This LED illuminates when there is proper arc transfer between the torch and the workpiece, and blinks during continuous pilot arc operation (such as when cutting expanded metal or moving the arc off the plate and then back on).
- **Start** – This LED illuminates when the power supply receives a start signal and remains illuminated during normal operation.

Figure 4



- | | |
|------------------|----------------|
| 1 Reset | 4 Start |
| 2 Error | 5 Ribbon cable |
| 3 Transfer (XFR) | |

During normal operation, the power-ON LED on the front of the power supply and the Start and Transfer LEDs on the control board illuminate. When a problem occurs with the system, one or more of the fault LEDs on the front of the power supply and the Error LED or the Reset LED on the control board may illuminate or blink.

Use the control board Reset and Error LEDs to troubleshoot

The Reset and Error LEDs provide information to use when troubleshooting a system failure. If the LEDs on the front of the power supply are blinking, count the number of times the Error LED blinks. Then, look at *Table 5* on page 77 to determine the corrective action.

Reset LED

When the control board's Reset LED illuminates, the voltages on the power board may be incorrect. Perform the following tests at J7 on the power board. See *Test 2 – power board voltage checks* on page 84.

- Test pin 5 to ground for 3.3 VDC ($\pm 10\%$).
- Test pin 7 to ground for 5 VDC ($\pm 10\%$).
- Test pin 12 to ground for 2.2 VDC ($\pm 10\%$).

If the values you find are not within $\pm 10\%$ of the three values listed above, detach the control board's ribbon cable and perform the tests again. If you find the correct values the second time, replace the control board. Otherwise, replace the power board. See *Replace the control board* on page 126 or *Replace the power board* on page 135.

Error LED

The number of times the Error LED blinks indicates the problem detected. Each blink is a half-second long, and each series of blinks is separated by a 2-second pause. See *System tests* (below) for detailed test procedures.

Table 5

Number of blinks	Problem	Solution
3	Faulty power board or control board	- Perform <i>Test 3 – VBUS and voltage balance on power board</i> on page 86. If any of the values are incorrect, replace the power board. - Perform <i>Test 2 – power board voltage checks</i> on page 84. If any of the values for pins 5, 7, or 12 are incorrect, remove the control board and test again. If the values are correct, replace the control board. - If the values for pins 5, 7, or 12 are correct when you perform <i>Test 2 – power board voltage checks</i> on page 84 but any other values from Test 2 are incorrect, replace the power board.
4	Faulty fan, solenoid valve, or power board	Perform <i>Test 4 – solenoid valve</i> on page 89 and <i>Test 8 – fan</i> on page 95. If the solenoid valve test and the fan test both pass, replace the power board. If Test 4 fails, replace the solenoid valve; if Test 8 fails, replace the fan.
6	Inverter saturation	Install new consumables in the torch. If you continue to see this error code, replace the power board.

System tests

You can use either the ground clip near the top of the rear panel ① or the ground screw on the heatsink ② (marked by the ground symbol on the power board) for any tests that require the multimeter to be attached to ground.

Figure 5 – Grounding options

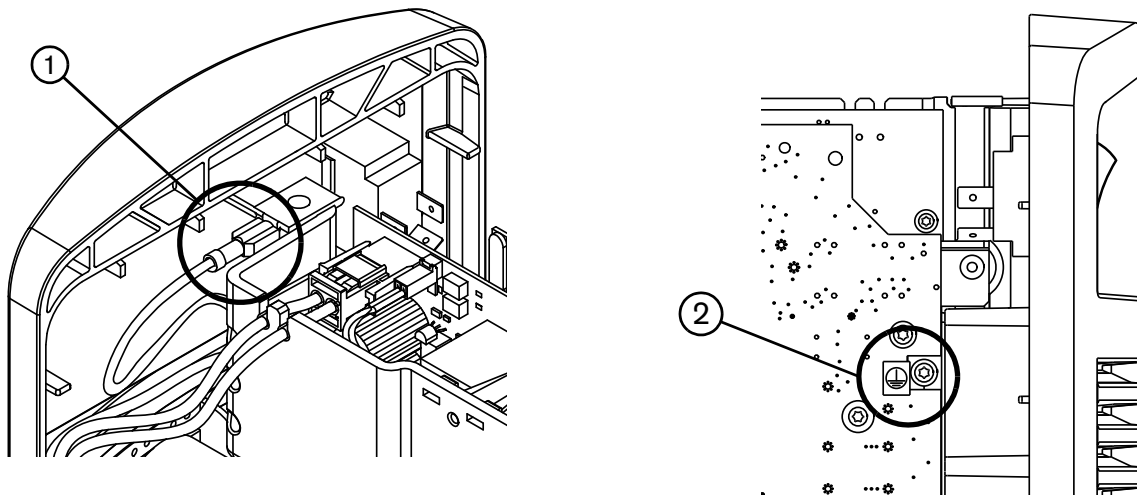
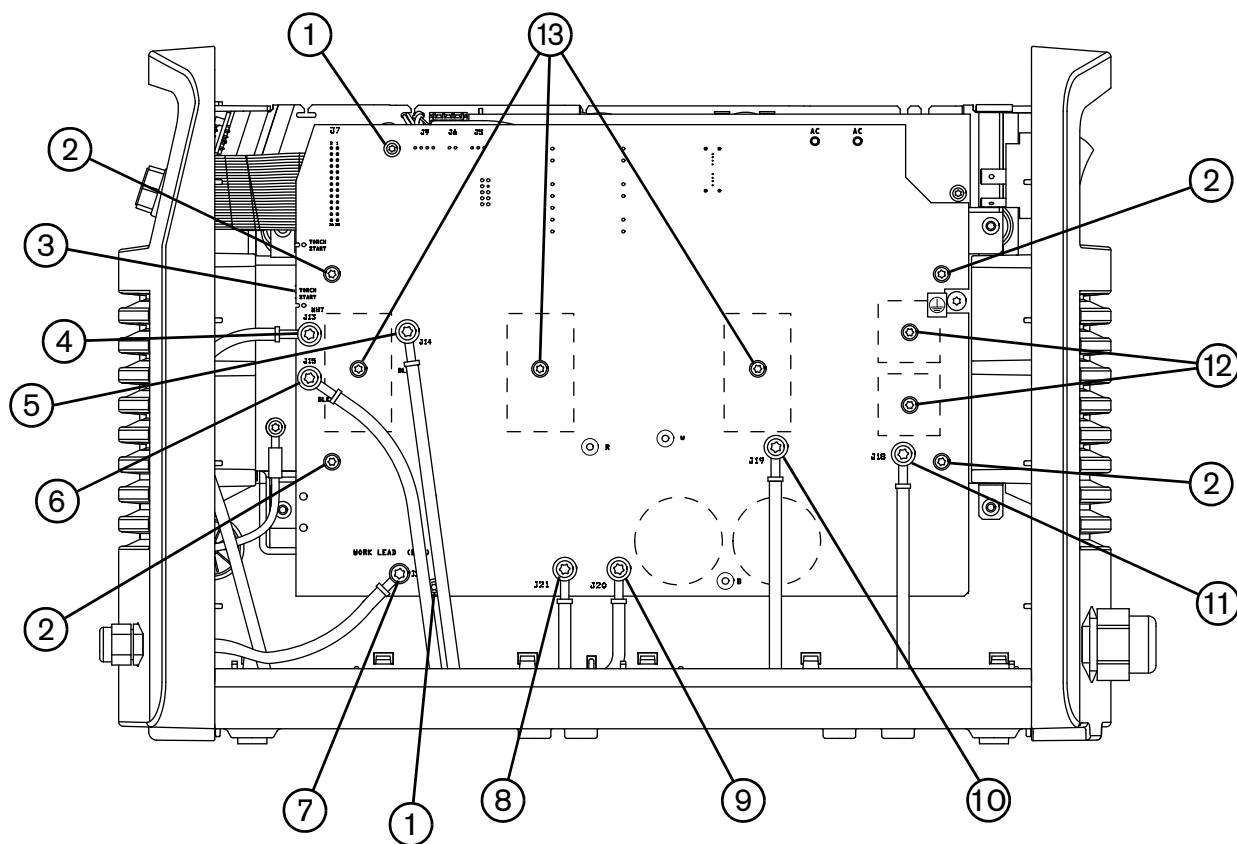


Figure 6 – Back of the power board



- | | |
|------------------------------------|----------------------------------|
| 1 Retaining screw (2) | 8 J21 |
| 2 Heatsink assembly screw (4) | 9 J20 |
| 3 Torch start and cap-sensor (J12) | 10 J19 |
| 4 J13 | 11 J18 |
| 5 J14 | 12 Input diode bridge screws (2) |
| 6 J15 | 13 IGBT screws (3) |
| 7 Work lead connector (J22) | |

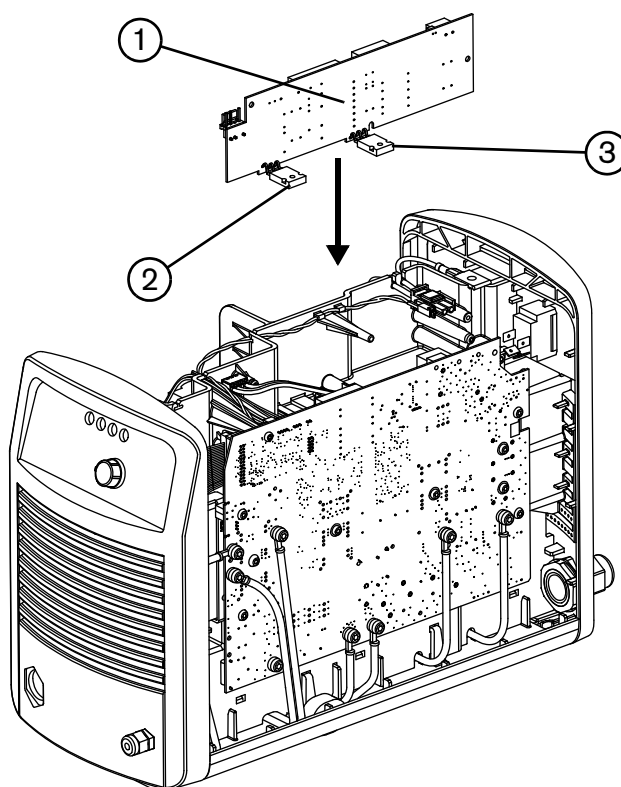


WARNING!

Voltages of up to 50 VDC continue to be present on the DC bus for at least 30 seconds after disconnecting the input power. Allow bus voltages to dissipate before performing any tests.

- Powermax30 AIR**
- Service Manual 808850

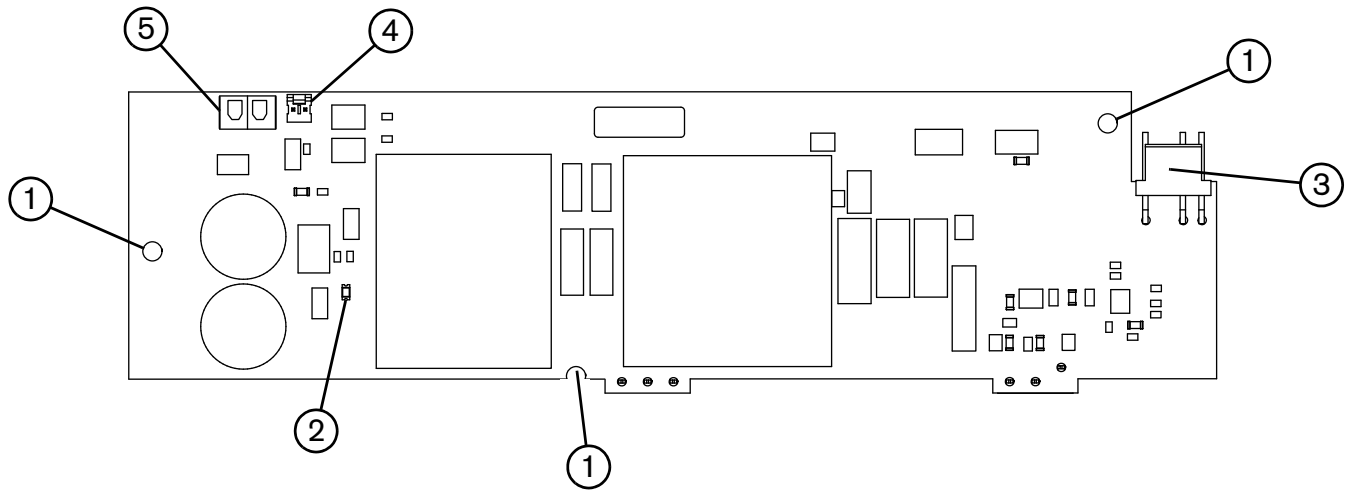
Figure 8 – Back of the compressor-driver board



- 1 Compressor-driver board
- 2 MOSFET

- 3 Diode

Figure 9 – Component side of the compressor-driver board



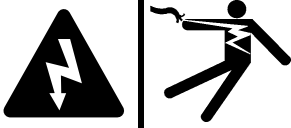
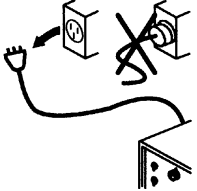
- 1 Retaining screw (3)
- 2 Diagnostic LED (D5)
- 3 J3

- 4 J1
- 5 J2

Test 1 – voltage input

Check the incoming voltage and the line voltage to the top of the power switch (S1).

- 1. Set power switch to OFF (O), and disconnect the power cord from the power source.
- 2. Remove the power supply handle and cover. See *Remove the power supply cover* on page 106.
- 3. Remove the component barrier. See *Remove the component barrier* on page 108.
- 4. Partially pull the power switch's top two wires out from the tabs and attach the multimeter test leads to the tabs to check the AC voltage. See *Figure 10* on page 83.

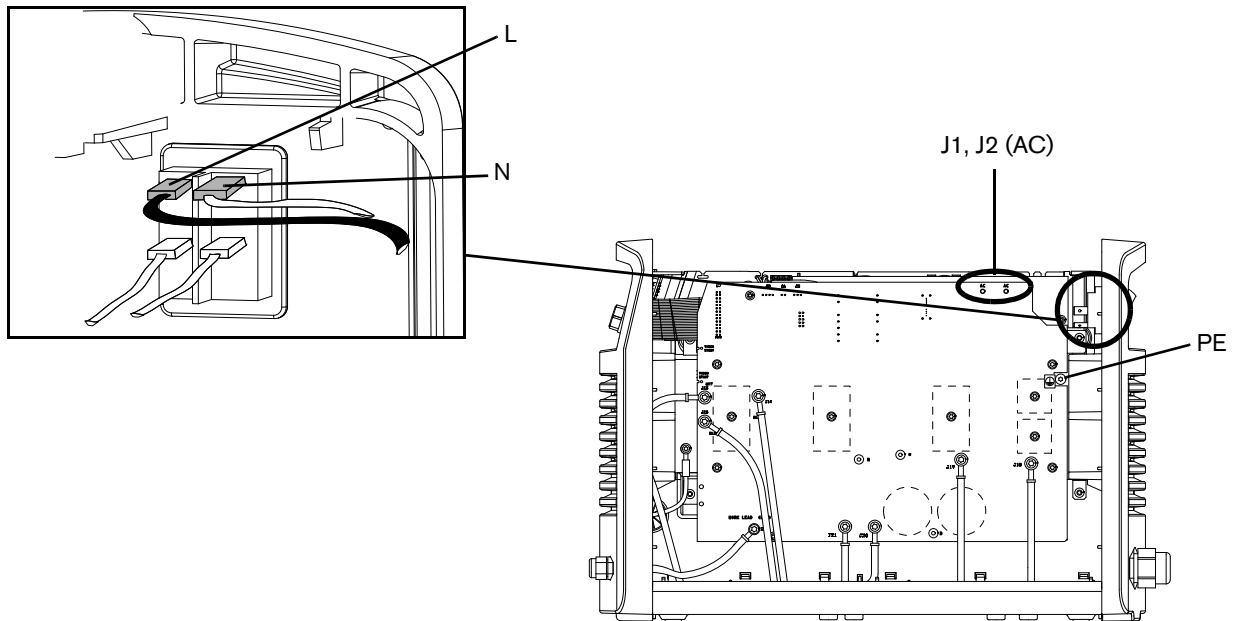
	WARNING! ELECTRIC SHOCK CAN KILL
	Use extreme caution when working near live electrical circuits. Dangerous voltages exist inside the power supply that can cause serious injury or death. See the <i>WARNING!</i> on page 59 before proceeding.

- 5. Once you have the test leads in place, leave the power switch set to OFF (O), and reconnect the electrical power. The voltage should equal the line voltage of the incoming circuit, for example 120 V or 240 V.



All values can be $\pm 15\%$.

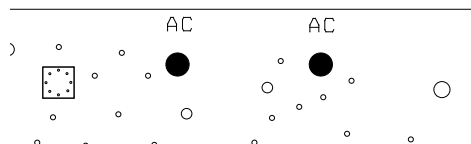
Figure 10



Single phase power supplies		
Designator	CSA wire colors	CE/CCC wire colors
L (live)	Black	Brown
N (neutral)	White	Blue
PE (ground)	Green	Green/yellow

6. If the AC voltage is incorrect, make sure you have power to the unit. If you do have power, inspect the power cord for damage, and replace if necessary. See *Replace the power cord and strain relief* on page 115.
7. If the power source and power cord are functioning correctly, disconnect the power cord again and reconnect the two wires to the power switch.
8. Reconnect the electrical power and set the power switch to ON (I).
9. Measure the AC voltage from J1 to J2 (labeled “AC” on the back of the power board). This value should be the same as the incoming line voltage. If it is not, check the power switch and replace if necessary. See *Replace the power switch* on page 124.

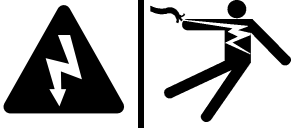
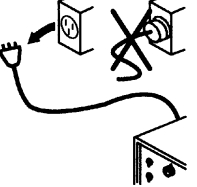
Figure 11



10. If the power-ON LED is still illuminated, perform *Test 2 – power board voltage checks* on page 84.

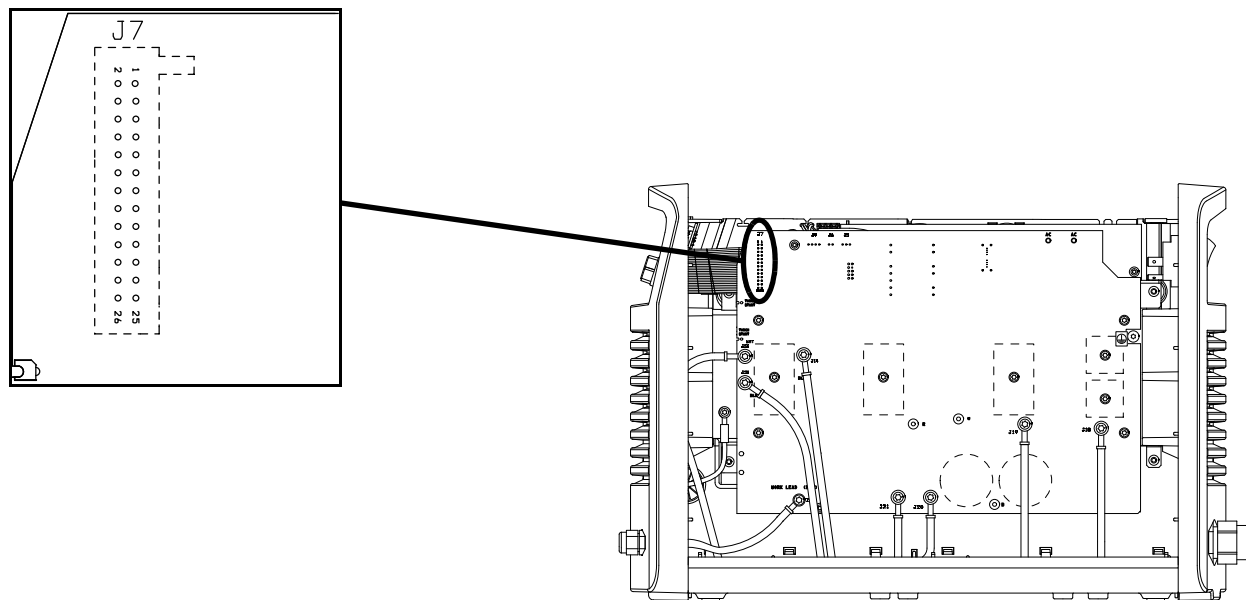
Test 2 – power board voltage checks

1. Set the power switch to OFF (O), and disconnect the power cord from the power source.
2. Remove the power supply handle and cover. See *Remove the power supply cover* on page 106.
3. Remove the component barrier. See *Remove the component barrier* on page 108.

	WARNING! ELECTRIC SHOCK CAN KILL
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4. Reconnect the electrical power and set the power switch to ON (I).
5. Use a multimeter to verify the voltages at the J7 pins listed in *Table 6* on page 85 to make sure the power board (PCB2) is functioning correctly. If any of the values are incorrect, replace the power board. See *Replace the power board* on page 135.

Figure 12



To test the values at pin 16, you must position the torch and power supply so that you can safely pull and release the torch trigger. For ground locations, see *Figure 5* on page 77.

**CAUTION!**

Do not use -VBUS (test point *W*) as ground. Doing so could destroy the power supply. Instead ground to either the ground wire clip on the rear panel or to the heatsink. See *Figure 5* on page 77.

Table 6

J7 pin number to ground	Test	Expected value
19	VACR (rectified AC line voltage)	0.86 V at 120 line voltage 1.87 V at 230 line voltage
21	VBUS (DC bus voltage)	2.28 VDC at 375 VBUS
18	IPFC (input current)	< 0.1 VDC
20	IFB (output current)	< 0.1 VDC
22	ITF (transfer current)	< 0.1 VDC
5	3.3 VDC	3.3 VDC $\pm$ 10%
7	5 VDC	5 VDC $\pm$ 10%
12	24 V sense pin	2.2 VDC $\pm$ 10%
16	Start signal	3.2 VDC closed 0 VDC open

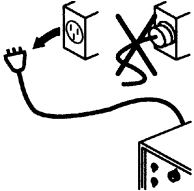


All values can be $\pm$ 10%.

Test 3 – VBUS and voltage balance on power board

Test the power board (PCB2) to make sure the circuits are balanced. The test points are labeled on the back of the power board, as are the voltages and positive and negative capacitor terminals. See *Figure 13* on page 88 for locations of test points.

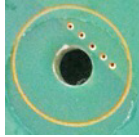
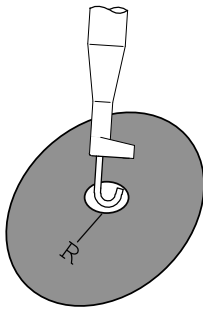
- 1. Set the power switch to OFF (O), and disconnect the power cord from the power source.
- 2. Remove the power supply handle and cover. See *Remove the power supply cover* on page 106.
- 3. Remove the component barrier. See *Remove the component barrier* on page 108.)

	WARNING! ELECTRIC SHOCK CAN KILL
	Use extreme caution when working near live electrical circuits. Dangerous voltages exist inside the power supply that can cause serious injury or death. See the WARNING! on page 59 before proceeding.

- 4. Reconnect the electrical power.
- 5. Use *Table 7* to identify how to attach to the test points on the power board based on the power board design in your system. The test points are labeled **R**, **W**, and **B** on all power boards.

	CAUTION!
Use only the test point types identified in <i>Table 7</i>. Follow the instructions in this procedure to correctly attach to the test point openings in the power board. Do not use a multimeter with standard test leads for this test. This can cause a short-circuit between the BUS and the heatsink. Do not use -VBUS (test point <i>W</i>) as ground. Doing so could destroy the power supply.	

Table 7 – Identify test point type on power board

Current power board design (CSA models: revision E and later) (CE/CCC models: revision D and later)	Prior power board design (CSA models: revision D and earlier) (CE/CCC models: revision C and earlier)
Each test point lined with copper ring 	No metal ring around test points 
Use miniature banana plugs* (Pamona® 2945 plugs or similar) 	Use test hooks 
* If miniature banana plugs are not available, use small test clips that you can attach to the copper contacts in the test point openings on the power board.	

6. Measure the voltage from **W** to **R**. See *Figure 13*.

- Position the banana plugs or test hooks at **W** and **R** on the power board.
- Turn the power ON (**I**).
- The multimeter should read 375 VDC.

If you get a value other than 375 VDC, multiply the reading by 0.00601 to convert it to millivolts. Test pin 21 on J7. See *Test 2 – power board voltage checks* on page 84. If the values match, it is a normal reading.

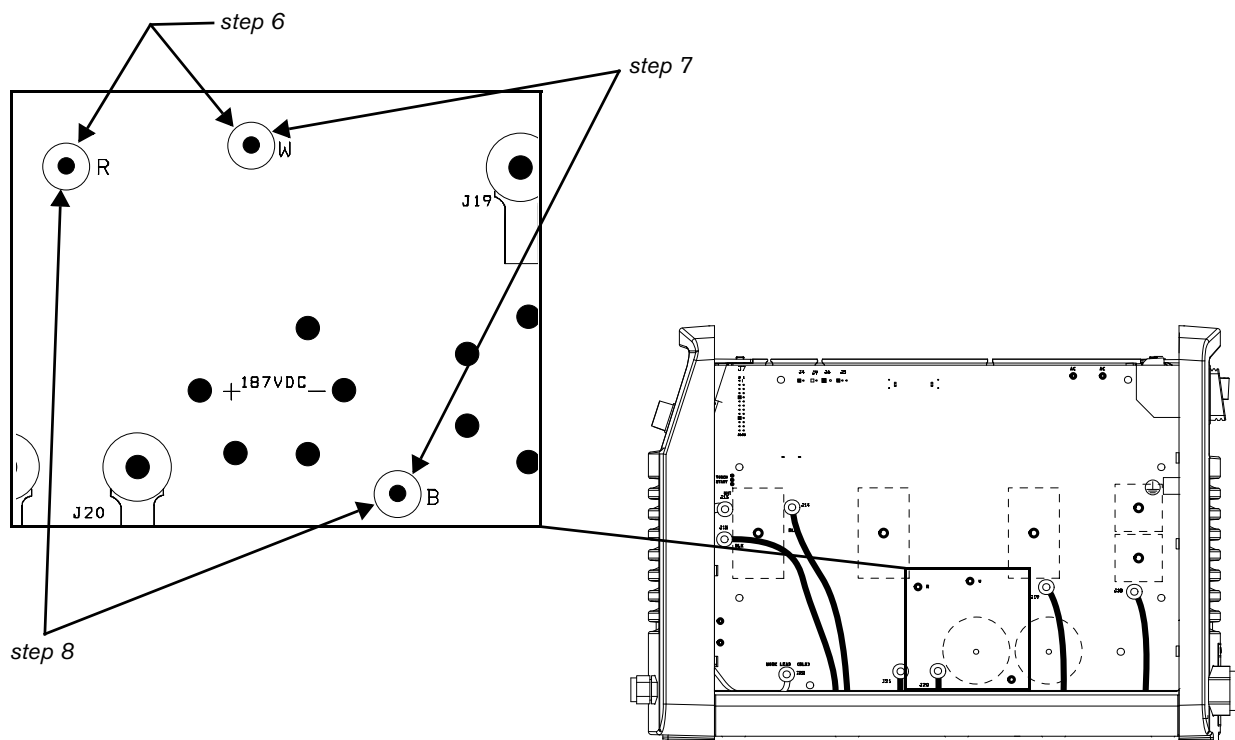
7. Measure the voltage from **W** to **B**.

- Turn the power OFF (**O**).
- Move the banana plugs or test hooks to **W** and **B**.
- Turn the power ON (**I**).
- This value should be 187.5 VDC or one-half of the value found in *step 6*.

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8. Measure the voltage from **R** to **B**.
 - a. Turn the power OFF (**O**).
 - b. Move the banana plugs or test hooks to **R** and **B**.
 - c. Turn the power ON (**I**).
 - d. This value should be 187.5 VDC or one-half of the value found in *step 6*.
9. The values found in *step 7* and *step 8* should be approximately equal. If they differ by more than 30 V, replace the power board.

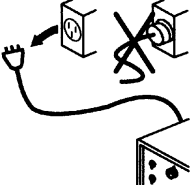
Figure 13



Test 4 – solenoid valve

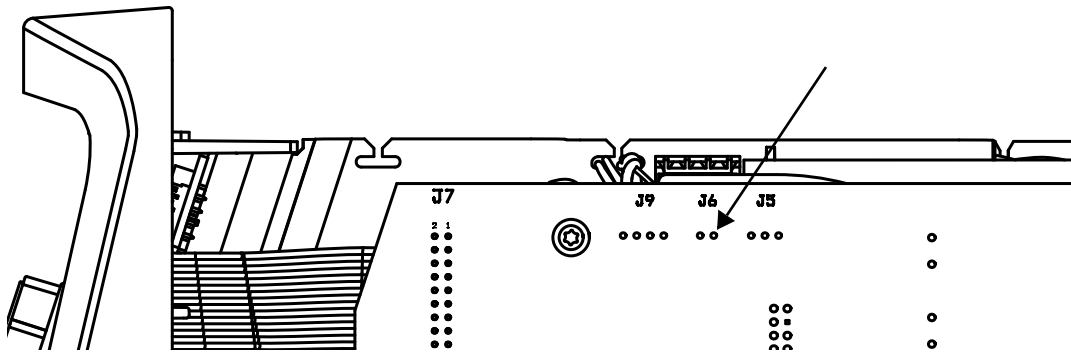
This test verifies the proper operation of the solenoid valve (V1).

1. Set the power switch to OFF (O), and disconnect the power cord from the power source.
2. Remove the power supply handle and cover. See *Remove the power supply cover* on page 106.
3. Remove the component barrier. See *Remove the component barrier* on page 108.

	WARNING! ELECTRIC SHOCK CAN KILL
	Use extreme caution when working near live electrical circuits. Dangerous voltages exist inside the power supply that can cause serious injury or death. See the <i>WARNING!</i> on page 59 before proceeding.

4. Reconnect the electrical power.
5. Place a jumper from pin 2 of J6 (black wire) on the power board to ground. See *Figure 5* on page 77 for grounding options.

Figure 14



6. Turn the power ON (I). The valve should click.
7. Remove the valve's connector from J6 on the power board.
8. Measure the voltage between pin 1 of J6 (red wire) and ground.
9. If you do not hear the valve click and the voltage check reads 24 VDC, replace the solenoid valve. See *Replace the solenoid valve* on page 151.

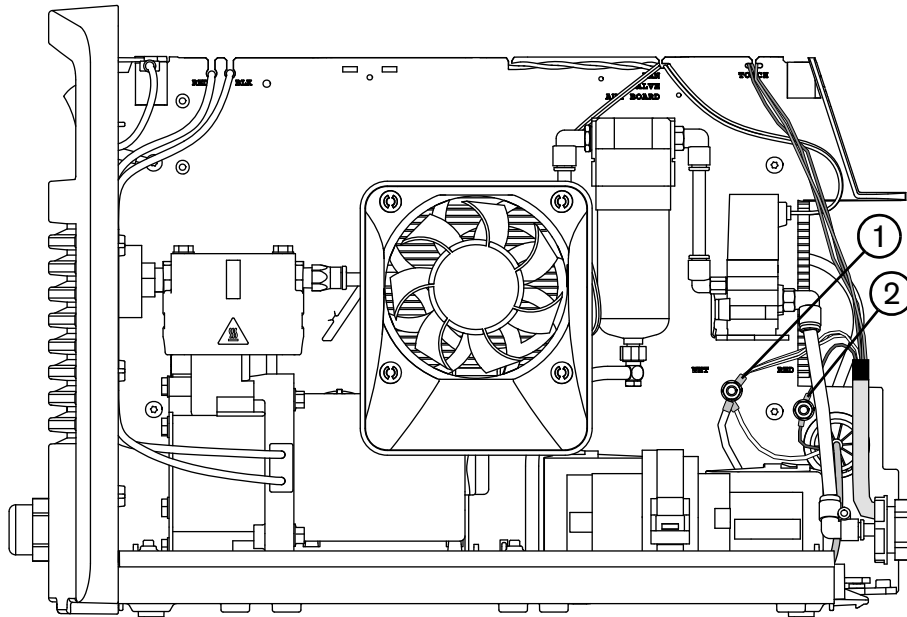
Test 5 – torch stuck open or torch stuck closed

If the nozzle and electrode are not in contact before the torch trigger is pulled, the power supply detects a “torch stuck open” fault. If the nozzle and electrode remain in contact after the torch trigger is pulled, the power supply detects a “torch stuck closed” fault.

Use the following test to determine if the torch is stuck in either position.

1. Set the power switch to OFF (O), and disconnect the power cord from the power source.
2. Remove the power supply handle and cover. See *Remove the power supply cover* on page 106.
3. Remove the component barrier. See *Remove the component barrier* on page 108.
4. Measure the resistance between the torch lead's white wire group ① and red wire ②. The resistance should read very low, a closed circuit.
 - ❑ If the resistance reads as open with the power OFF, the nozzle and electrode are not in contact, or one of the wires in the lead is broken. Make sure that the torch plunger moves freely in the torch head.

Figure 15



The front panel is hidden in this figure to better show the torch lead and wires.



CAUTION!

To avoid causing a short or damage to the multimeter, do not fire the torch with the multimeter connected to the torch lead's white wire group and red wire while the system is powered ON.

5. Turn ON (I) the power. With the system idle, quickly tap the torch trigger to force gas flow without firing an arc. Is gas flowing?
 - ❑ If yes, the gas will flow for 5 seconds before shutting off. While the gas is flowing, measure the resistance between the torch lead's white wire group ① and red wire ②. The resistance should read very high – an open circuit – when gas is flowing.

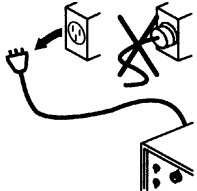
If the resistance reads as very low (closed circuit) with the gas flowing, the nozzle and electrode are in contact or a short-circuit occurred in one of the wires in the torch lead. Continue with *step 6*.
 - ❑ If no, the solenoid valve may be faulty. Perform *Test 4 – solenoid valve* on page 89.
6. Does the torch plunger move freely in the torch head?
 - ❑ If yes, replace the torch lead. See *Replace the torch lead* on page 205.
 - ❑ If no, replace the torch body. See *Replace the torch body* on page 201.
7. Because “torch stuck open” and “torch stuck closed” failures can be intermittent, repeat the test several times.

Test 6 – plasma start

Use this test to make sure the control board LED is receiving a valid start signal.

- 1. Set the power switch to OFF (O), and disconnect the power cord from the power source.
- 2. Remove the power supply handle and cover. See *Remove the power supply cover* on page 106.
- 3. Remove the component barrier. See *Remove the component barrier* on page 108.

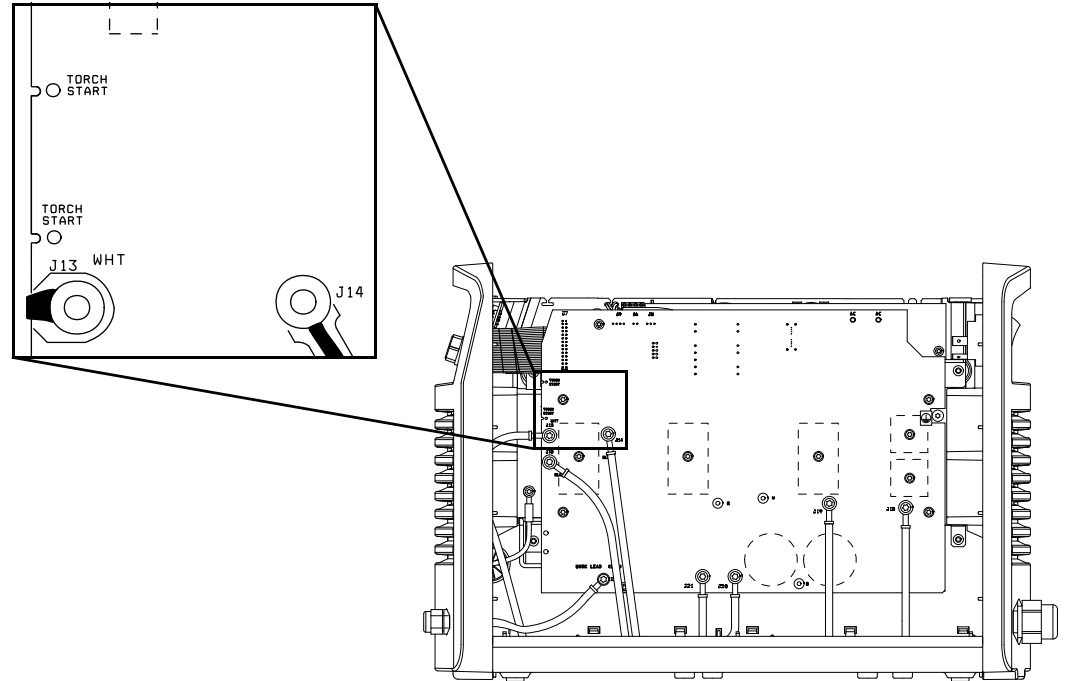
		WARNING!
	While testing, do not touch the nozzle at the end of the torch. Dangerous voltages can cause serious injury.	

		WARNING! ELECTRIC SHOCK CAN KILL
		Use extreme caution when working near live electrical circuits. Dangerous voltages exist inside the power supply that can cause serious injury or death. See the <i>WARNING!</i> on page 59 before proceeding.

- 4. Reconnect the electrical power and turn the power ON (I).
- 5. Quickly tap the torch trigger to force gas flow without firing an arc, and watch the Start LED on the control board. The Start LED should illuminate whenever the trigger is pulled. See *Control board LEDs* on page 75 for the location of the Start LED.
- 6. Turn OFF (O) the power.

7. Check the resistance at the 2 torch-start test points on the power board. See *Figure 16*.
 - a. With the trigger pulled, the resistance should read 10 Ω or less.
 - b. With the trigger released, the circuit should read approximately 3 k Ω .

Figure 16

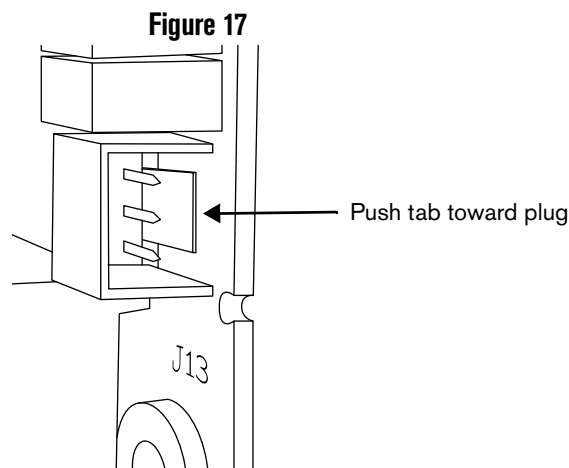


8. If this test fails, check the torch start switch and the torch wires. Replace if necessary. See *Replace the start switch* on page 203 or *Replace the torch lead and strain relief* on page 153.
9. Turn ON (I) the power.
10. Measure pin 16 of J7 to ground. See *Test 2 – power board voltage checks* on page 84.
 - a. With the trigger pulled, it should measure as 0 VDC for an open circuit.
 - b. With the trigger released, it should measure 3.2 VDC for a closed circuit.
11. Are the values correct?
 - ❑ If yes, replace the control board. See *Replace the control board* on page 126.
 - ❑ If no, replace the power board. See *Replace the power board* on page 135.

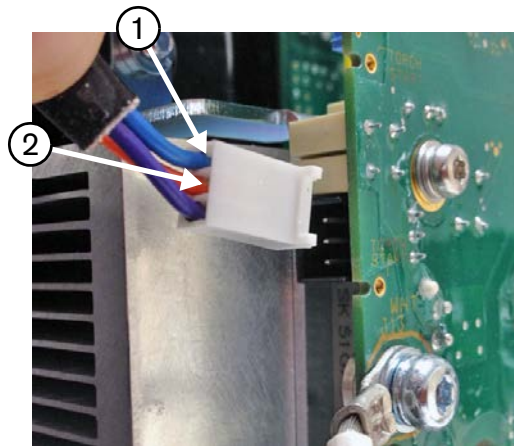
Test 7 – torch cap-sensor

Test the torch cap-sensor switch and torch leads.

1. Set the power switch to OFF (O), and disconnect the power cord from the power source.
2. Remove the power supply handle and cover. See *Remove the power supply cover* on page 106.
3. Remove the component barrier. See *Remove the component barrier* on page 108.
4. Detach the front panel. See *Detach the front panel* on page 110.
5. Disconnect the cap-sensor connector at J12 on the heatsink side of the power board by pushing the tab on the connector toward the plug and pulling the plug out.



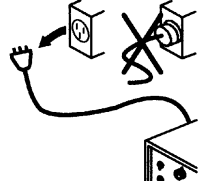
6. Measure the resistance from the blue wire (pin 1) ① to the orange wire (pin 2) ②. It should measure less than 10 Ω . If it measures high resistance, the cap-sensor switch circuit is open.
7. Make sure the torch plunger moves smoothly. If it does not, replace the torch body. See *Replace the torch body* on page 201.
8. Make sure the consumables are correctly installed. Adjust the consumables if necessary.
9. If the torch parts mentioned in *step 7* and *step 8* are working properly, the cap-sensor switch is faulty or the torch lead has a broken wire. Replace the faulty part. See *Replace the cap-sensor switch* on page 204 or *Replace the torch lead and strain relief* on page 153.



Test 8 – fan

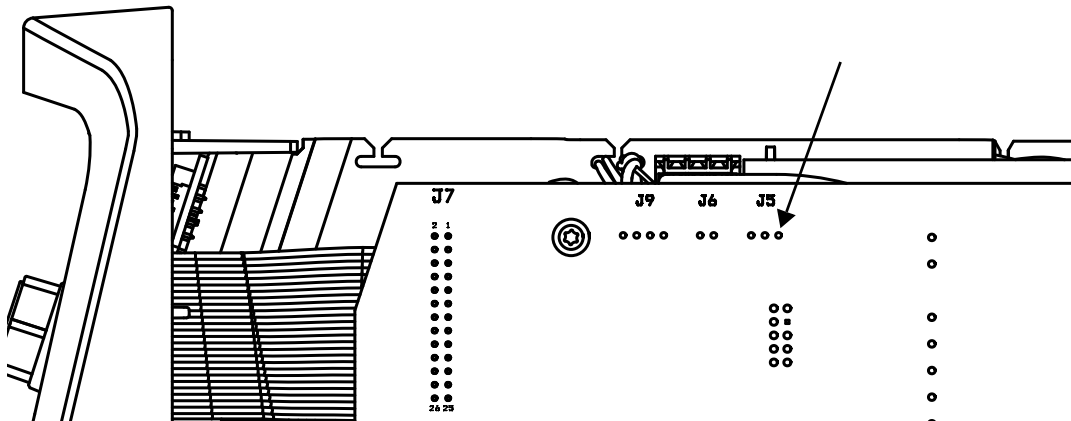
Test the fan (M1) for proper operation.

1. Set the power switch to OFF (O), and disconnect the power cord from the power source.
2. Remove the power supply handle and cover. See *Remove the power supply cover* on page 106.
3. Remove the component barrier. See *Remove the component barrier* on page 108.
4. Place a jumper from ground (see *Figure 5* on page 77 for grounding options) to pin 3 of J5 (black wire, see *Figure 18*).

		WARNING! ELECTRIC SHOCK CAN KILL
	Use extreme caution when working near live electrical circuits. Dangerous voltages exist inside the power supply that can cause serious injury or death. See the <i>WARNING!</i> on page 59 before proceeding.	

5. Reconnect the electrical power and set the power switch to ON (I).
6. If the fan does not operate, replace the fan. See *Replace the fan* on page 164.

Figure 18



Because of protection features on the fan driver chip, this test can trigger a fault. You can disregard this fault if it occurs as a result of a fan test. The purpose of the fan test is to verify that the fan is operating properly, not to test the fan drive circuit.

Test 9 – internal compressor, compressor-driver board, and power board

If the internal compressor fault LED and temperature fault LED both illuminate, the root cause may be:

- Faulty internal compressor
- Faulty compressor-driver board (PCB3)
- Faulty power board (PCB2)

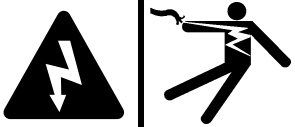
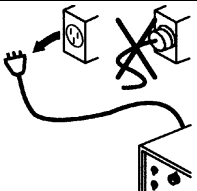
Use this test to systematically check each of these 3 components to determine which one needs to be replaced.



For a high-level overview of this procedure, see *Process flow for troubleshooting internal compressor and temperature fault LEDs* on page 228.

Turn OFF the power and remove the cover

1. Set the power switch to OFF (O), and disconnect the power cord from the power source.
2. Remove the power supply handle and cover. See *Remove the power supply cover* on page 106.
3. Remove the component barrier. See *Remove the component barrier* on page 108.)

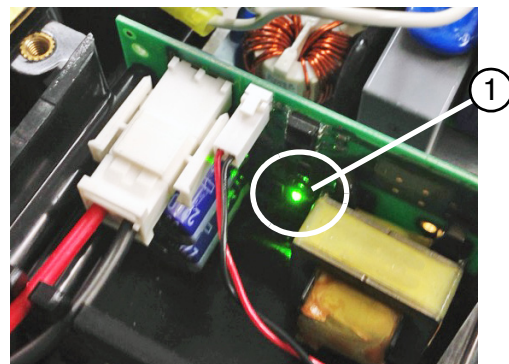
	WARNING! ELECTRIC SHOCK CAN KILL
	Use extreme caution when working near live electrical circuits. Dangerous voltages exist inside the power supply that can cause serious injury or death. See the <i>WARNING!</i> on page 59 before proceeding.

Test 9a – Check the diagnostic LED (D5) on the compressor-driver board

4. Reconnect the power cord, and set the power switch to ON (I).
5. Quickly tap the torch trigger to force gas flow without firing an arc, then watch the green diagnostic LED (D5) ① on the compressor-driver board. Does the LED illuminate while the torch is being fired?
 - a. If yes, check the internal compressor. Skip to *step 10*.
 - b. If no, continue with the next step.

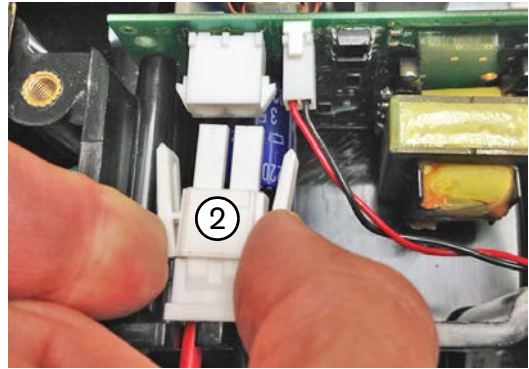


The D5 diagnostic LED indicates whether or not there is 15 VDC output from the compressor-driver board to the internal compressor.



6. Set the power switch to OFF (O).

7. Remove the J2 connector ② from the compressor-driver board.
8. Set the power switch to ON (I).
9. Quickly tap the torch trigger, then watch the green diagnostic LED (D5) on the compressor-driver board. Does the LED illuminate while the torch is being fired?
 - a. If yes, check the internal compressor. Continue with the next step.
 - b. If no, check the compressor-driver board. Skip to step 16.



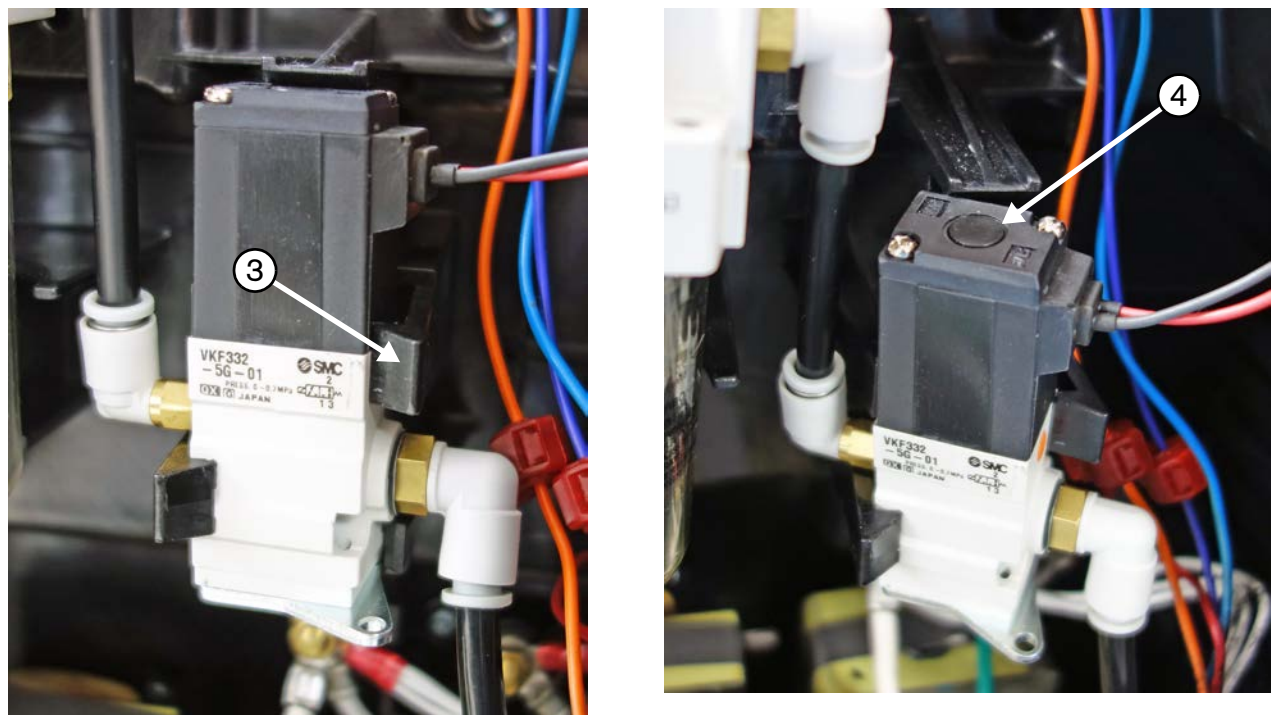
Test 9b – Check the internal compressor

		WARNING!
	While testing, do not touch the nozzle at the end of the torch. Dangerous voltages can cause serious injury.	

		WARNING! HOT SURFACE CAN CAUSE INJURY
	Allow the internal compressor to cool before touching it. The surface of the compressor can become very hot during use and can cause serious injury if it comes in contact with skin.	

10. Set the power switch to OFF (O).
11. On the fan side of the power supply, pull back the clip on the right side of the solenoid valve, and work the valve free from that clip ③. Carefully tilt the top of the valve towards you until you can see the gas shut-off override button ④. See Figure 19.
12. Connect the internal compressor to an external 12 VDC power source (for example, a car battery).
13. Hold down the gas shut-off override button on top of the solenoid valve, then quickly tap the torch trigger. Is the compressor running, and is air blowing from the torch?
 - a. If yes, check the compressor-driver board. Continue with the next step.
 - b. **If no, replace the internal compressor.** See *Replace the internal compressor* on page 169. Be sure to push the solenoid valve back into place between both clips from the center panel.

Figure 19

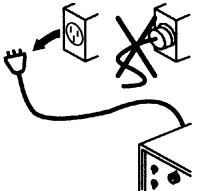


- 14. Push the solenoid valve back into place between both clips from the center panel.
- 15. Make sure both of the solenoid valve's gas supply hoses are connected securely to their fittings.

Test 9c – Check the compressor-driver board

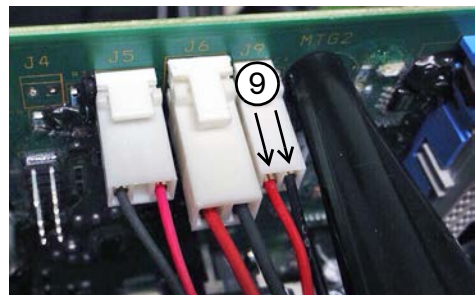
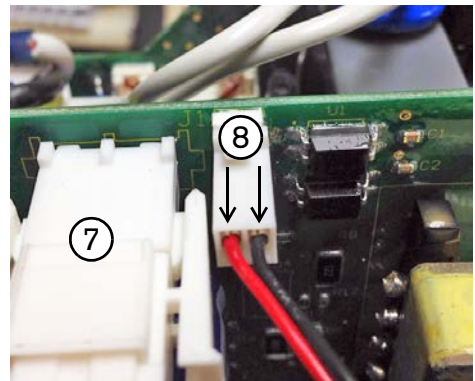
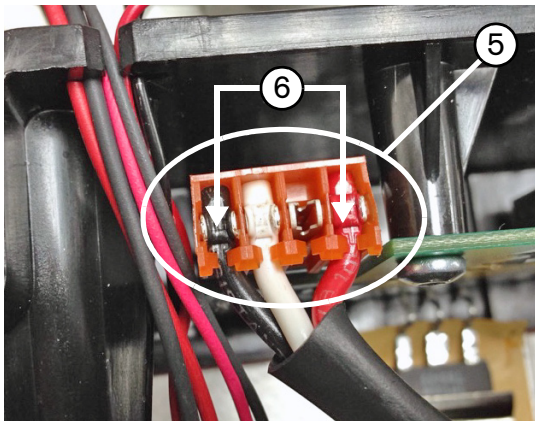
- 16. Set the power switch to OFF (O).
- 17. Remove the white cap from the top of the J3 connector ⑤ on the compressor-driver board. See Figure 20 on page 99.

 You may need to unplug the J3 connector in order to remove the white cap. If you do, plug the connector back in before continuing with the next step.

	WARNING! ELECTRIC SHOCK CAN KILL
	Use extreme caution when working near live electrical circuits. Dangerous voltages exist inside the power supply that can cause serious injury or death. See the <i>WARNING!</i> on page 59 before proceeding.

18. Set the power switch to ON (I).
19. Check VBUS on the red and black wires on the J3 connector (pin 1 and pin 4) ⑥. Is VBUS present (nominal 375 VDC bus voltage)?
 - a. If yes, check the compressor-enable voltage on the compressor-driver board. Skip to *step 21*.
 - b. If no, remove the J2 connector ⑦ from the compressor-driver board. Continue with the next step.
20. Repeat the previous step to check VBUS with J2 disconnected. Is VBUS present (nominal 375 VDC bus voltage)?
 - a. If yes, check the compressor-enable voltage on the compressor-driver board. Continue with the next step.
 - b. If no, check the power board. Skip to *step 23*.
21. Quickly tap the torch trigger, then check the compressor-enable voltage on the J1 connector ⑧ on the compressor-driver board. (The J1 connector may look different from the one shown in *Figure 20*.) Is compressor-enable voltage present (3 VDC)?
 - a. **If yes, replace the compressor-driver board.** See *Replace the compressor-driver board* on page 129.
 - b. If no, check the twisted pair wires that connect the compressor-driver board to the power board. Continue with the next step.
22. Quickly tap the torch trigger, then check the compressor-enable voltage on the J9 connector ⑨ on the power board. (The J9 connector may look different from the one shown in *Figure 20*.) Is compressor-enable voltage present (3 VDC)?
 - a. **If yes, replace the J1-to-J9 twisted pair wires.** See *Replace the wire group* on page 146.
 - b. If no, check the power board. Continue with the next step.

Figure 20

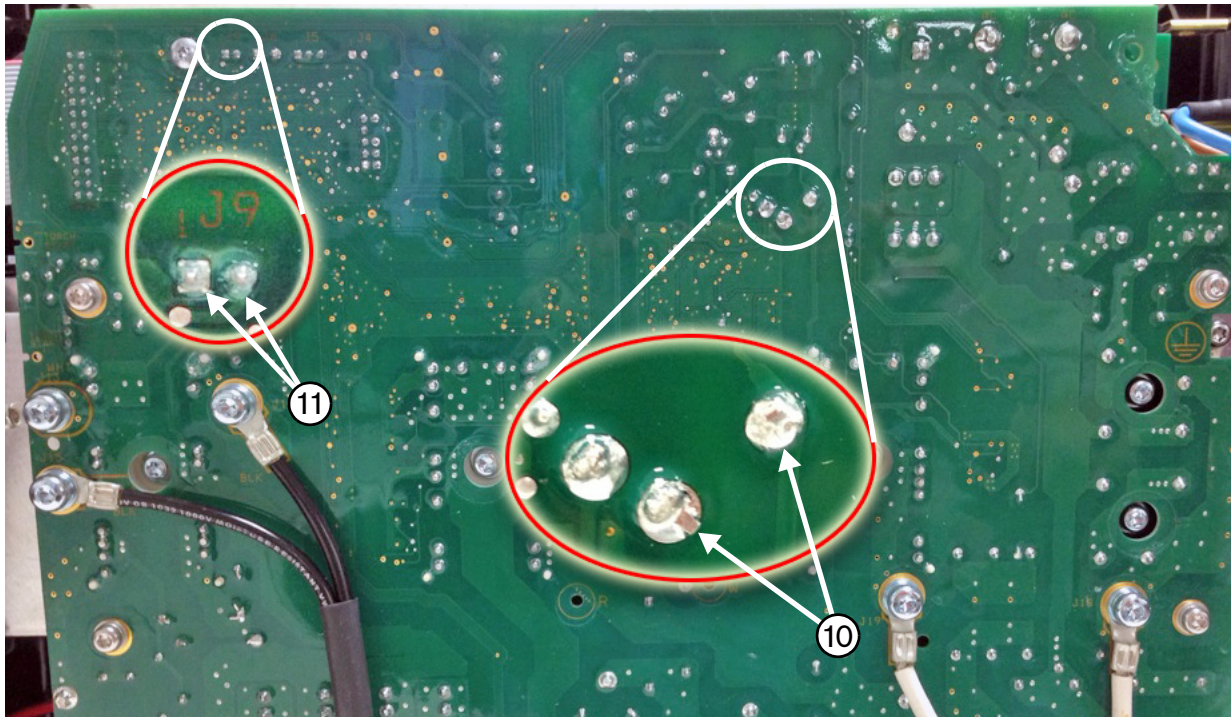


5 – Troubleshooting and System Tests

Test 9d – Check the power board

23. Set the power switch to OFF (O).
24. Remove the J1 and J3 connectors from the compressor-driver board. This isolates the power board.
25. Set the power switch to ON (I).
26. Check VBUS on the power board (nominal 375 VDC bus voltage) ⑩.
27. Quickly tap the torch trigger, then check compressor-enable voltage at J9 on the power board (3 VDC) ⑪.

Figure 21



28. Are VBUS and compressor-enable voltage present on the power board?
 - a. If VBUS and compressor-enable voltage are both present on the power board, **replace the compressor-driver board**. See *Replace the compressor-driver board* on page 129.
 - b. If neither VBUS nor compressor-enable voltage are present on the power board, **replace the power board**. See *Replace the power board* on page 135.
 - c. If VBUS or compressor-enable voltage is present on the power board but the other is not, **replace the power board**. See *Replace the power board* on page 135.

Test 10 – check air pressure

Use this test to determine if low air pressure is affecting system performance.

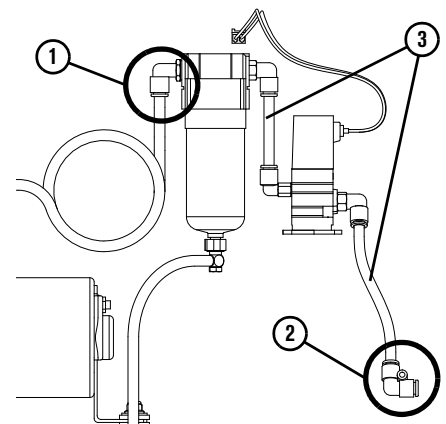
Tools required

- ❑ Air pressure test gauge kit (428643)
- ❑ Leak detector solution (for example, Snoop®) or a mixture of gentle soap and water
- ❑ Silicone lubricant (027055)
- ❑ Multimeter
- ❑ Assorted Phillips® and TORX® screwdrivers



Troubleshooting steps

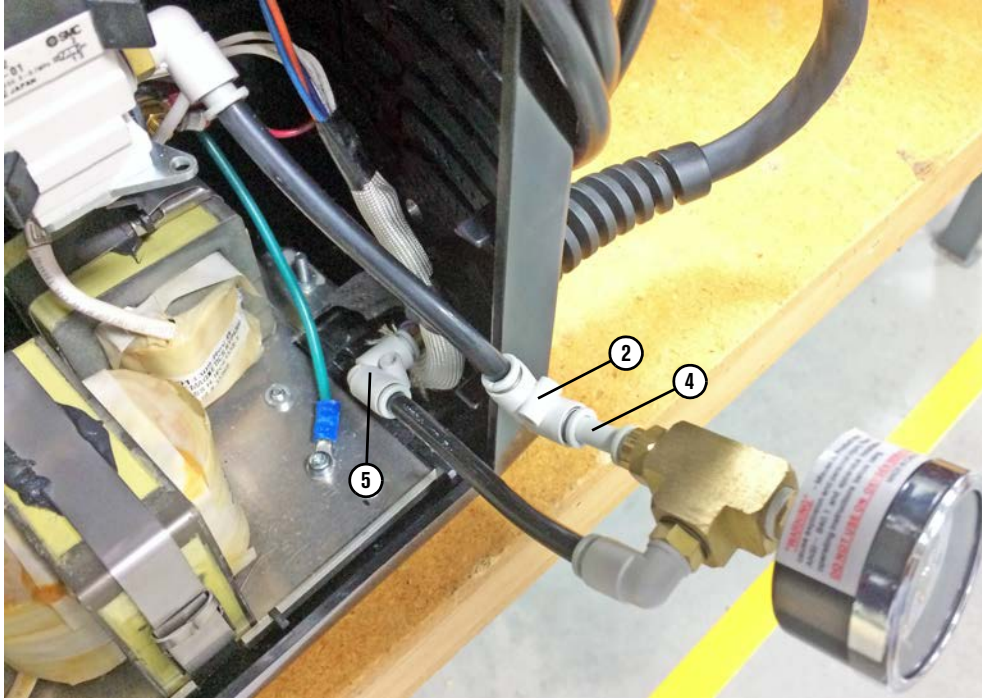
1. Set the power switch to OFF (O), and disconnect the power cord from the power source.
2. Check the O-ring on the torch head.
 - ❑ If the torch O-ring is dry, apply a thin film of silicone lubricant (027055) on the O-ring and the threads. The O-ring should look shiny, but there should not be any excess or built-up lubricant.
 - ❑ If the O-ring is cracked or worn, replace it (428179).
3. Install new consumables. Make sure they are the correct consumables for this system and torch and that they are installed correctly. See page 45.
4. Remove the power supply handle and cover. See page 106.
5. Make sure the heat exchange coil is fully seated in the 90° fitting ① on the air filter.
6. Make sure the 90° fitting ② is fully seated on the brass gas supply fitting from the torch lead.
7. Examine the 2 gas supply hoses ③ that connect to the solenoid valve. Make sure the hoses are fully seated in each 90° fitting. If the hoses are kinked or damaged, replace them. See page 142.

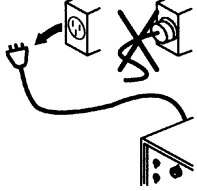


5 – Troubleshooting and System Tests

8. Connect the air pressure test gauge kit (428643) inside the power supply as shown in *Figure 22*:
- From the fan side of the power supply, push-to-release the 90° fitting ② from the torch lead gas supply fitting.
 - Push-to-connect the 90° fitting ② onto the short tube of the pressure test gauge assembly ④.
 - Push-to-connect the 90° fitting from the pressure test gauge ⑤ onto the torch lead gas supply fitting.

Figure 22 – Pressure test gauge connected to power supply



	WARNING! ELECTRIC SHOCK CAN KILL
	Use extreme caution when working near live electrical circuits. Dangerous voltages exist inside the power supply that can cause serious injury or death. While testing, do not touch the nozzle at the end of the torch. Dangerous voltages can cause serious injury. See the WARNING! on page 59 before proceeding.

9. Reconnect the power cord, and set the power switch to ON (I).
10. Set the cutting current (amperage) as follows:
- ☐ 22 A for 120 V electrical service
 - ☐ 30 A for 240 V electrical service

11. Perform a test cut in order to sustain a plasma arc for several seconds. Check the pressure reading on the gauge. It should read 2.9 – 3.4 bar (42 – 50 psi). (The pressure will be lower during postflow.)

With the compressor running, the needle on the gauge vibrates a lot. Ignore the highest and lowest points on the needle “swings.” Make sure that, on average, the pressure remains within the range specified above.

12. Repeat the previous step 2–3 times to make sure the pressure readings are consistent.
13. Set the power switch to OFF (O), and disconnect the power cord from the power source.
14. Remove the pressure test gauge assembly. Reconnect the 90° fitting to the gas supply fitting from the torch lead.
15. Reconnect the power cord, and set the power switch to ON (I).
16. If the air pressure was low, check the air compressor for proper voltage (15 V). See *Test 9a – Check the diagnostic LED (D5) on the compressor-driver board* on page 96.
17. Apply leak detector solution (for example, Snoop®) to check for leaks at the following points on the gas supply line. Quickly tap the torch trigger to force gas flow without firing an arc. See *Figure 23* on page 104.
- The 6 push-to-connect fittings between the air compressor and the torch lead gas supply fitting
 - The 2 gas hoses that connect to each side of the solenoid valve
 - Where the filter bowl screws into the air filter assembly
 - Where the drain hose connects to the bottom of the filter bowl

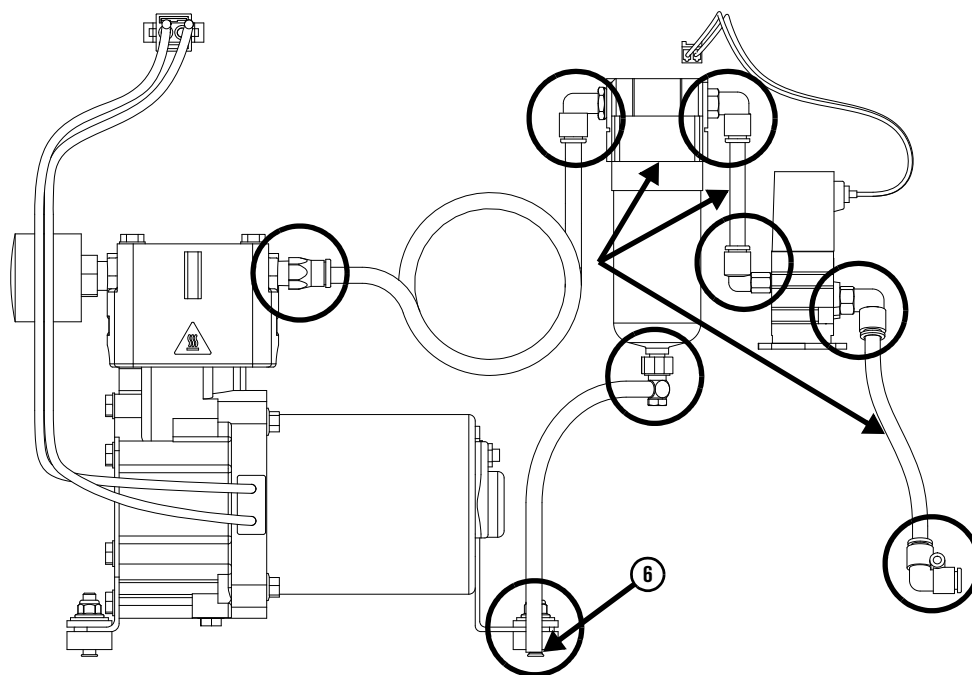


The fan may cause the appearance of a leak on some fittings even if there is not one. To avoid this, disconnect the fan from J5 on the power board before performing the leak test above. **Make sure to reconnect the fan before using the system.**

Replace damaged or defective parts as needed. See page 215 for replacement kit numbers.

18. Also check for leaks where the drain hose connects to the base of the power supply ⑥. Tilt the power supply so that you can feel if air is escaping through the hole in the base where the drain hose lets out.
19. Install a different torch to see if the low pressure issue persists. See page 153.
20. If low pressure persists after you complete all of these checks, the internal air compressor may be faulty. Install a new air compressor. See page 169.

Figure 23 – Where to check for gas leaks



Process flow for troubleshooting internal compressor and temperature fault LEDs

